

PROJECT 04

OTTUTHAL — Automated Bonding Tape Machine

MAS Kreeda Vaanavil (MAS Active Group)
2019 (3-month Internship) · Autonotation / Automation Engineering Intern

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PROJECT SUMMARY

Designed and built a semi-automated machine to apply heat-bonding tape onto garment panels at MAS Kreeda Vaanavil, replacing a four-step manual process (measure, cut, position, iron). The operator only places the garment and presses two start buttons. PLC-controlled PID temperature and pneumatic pressing ensure consistent bond quality. The project also produced research and CAD designs for a next-generation fully-automatic successor.

OBJECTIVES

- Automate tape feeding, measuring, cutting, and heat-bonding in one machine cycle.
- Let the operator set tape length for different garment styles.
- Hold bonding temperature within $\pm 5^{\circ}\text{C}$ using PID control.
- Two-hand start interlock — operator hands away from moving parts during cycle.
- Design recommendations for a next-generation fully-automatic machine.

HOW IT WORKS

Step 1: Stepper motor feeds tape from a roll; step count measures length; pneumatic guillotine cuts at setpoint. Step 2: Operator places garment and holds both start buttons; rodless cylinder moves garment under the heated pressing head; cartridge heater (PID-controlled via thermocouple + SSR) bonds the tape under pneumatic pressure. PLC ladder logic controls the full sequence and safety interlock.

KEY COMPONENTS

Component	Specification / Function
PLC (Haiwell / Xinje)	Sequence control, PID loop, and safety interlock
Cartridge Heater + Thermocouple	Bonding head heat source and temperature feedback
Solid State Relay (SSR)	Heater switching
Rodless Cylinder + Linear Rail	Fabric panel transport
Pneumatic Press Cylinder	Bonding head pressing action
Solenoid Valves (5/2-way)	Cylinder direction control
NEMA 17 Stepper + Driver	Tape feed and length measurement
Two-Hand Start Buttons	Category 2 safety interlock

TECHNICAL SPECIFICATIONS

Parameter	Value	Notes
Control	PLC ladder logic	Haiwell / Xinje
Temperature	PID, cartridge heater	$\pm 5^{\circ}\text{C}$ at setpoint
Tape drive	Stepper motor, step-count measurement	—
Safety	Two-hand start interlock	Both buttons held during cycle
Control power	24V DC	—

MY CONTRIBUTION

- Identified the manual bonding process as the key automation opportunity.
- Designed tape feed, cutting, fabric transport, and bonding head sub-systems.
- Wrote PLC ladder logic including PID loop and two-hand safety interlock.
- Ran experiment programme (tape length accuracy, bond temperature, cylinder timing).
- Produced CAD drawings and design report for recommended next-gen machine.

CHALLENGES & SOLUTIONS

Tape curling and static caused inconsistent positioning on the garment after cutting. Partly improved through guide geometry changes. This became the main driver for recommending an air-jet (Bernoulli effect) fabric-separation system for the next-generation machine.

RESULTS

Semi-automated machine delivered. Cycle time reduced vs. manual process. Bonding temperature held at setpoint. Two-hand safety interlock verified. Design report and CAD concepts for the next-generation fully-automatic machine submitted.

TOOLS & SOFTWARE USED

Tool / Software	Used For
SolidWorks	Machine design and concept drawings
Haiwell / Xinje PLC	Ladder logic and PID programming
Pneumatic Schematic	Circuit layout for actuator sequencing
Multimeter	Wiring and voltage checks

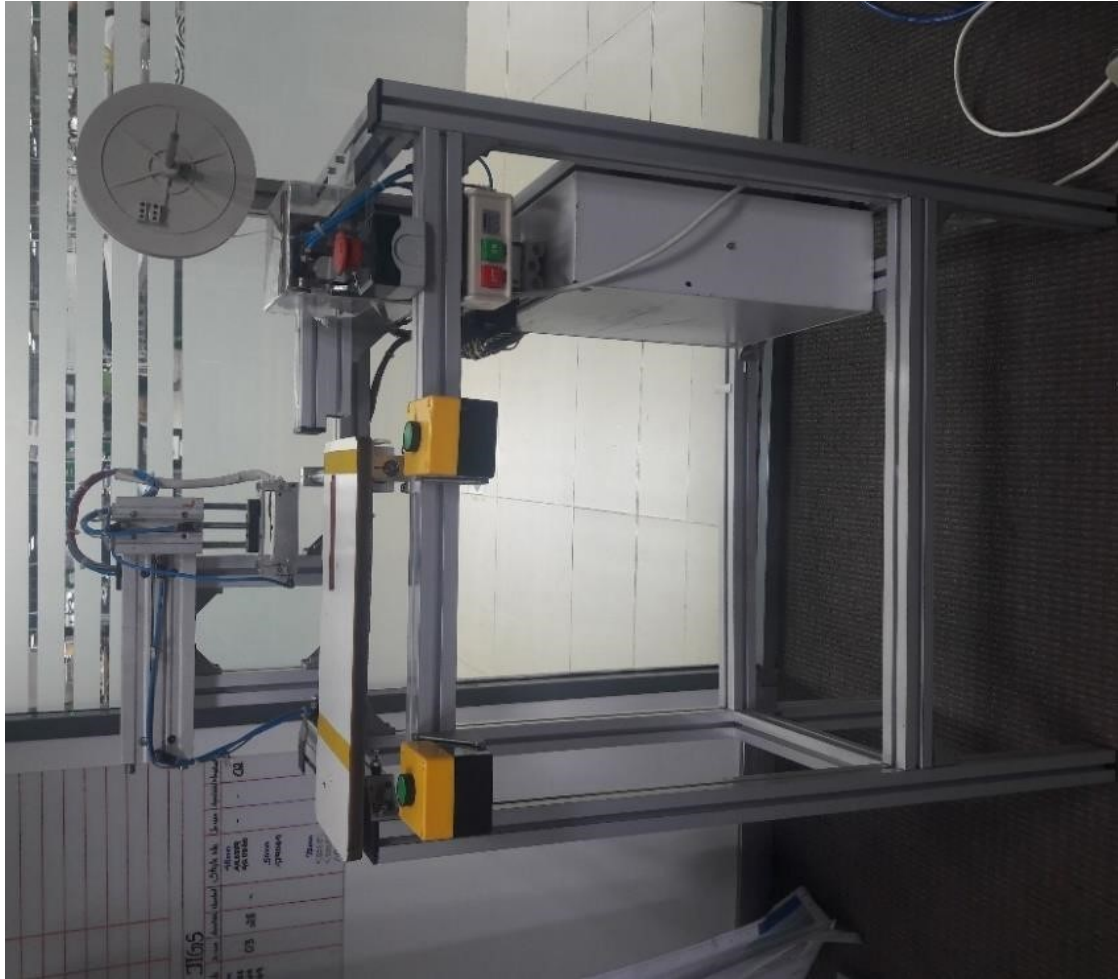
IMAGES & TECHNICAL DRAWINGS



Bonding tape — tape roll used in garment production



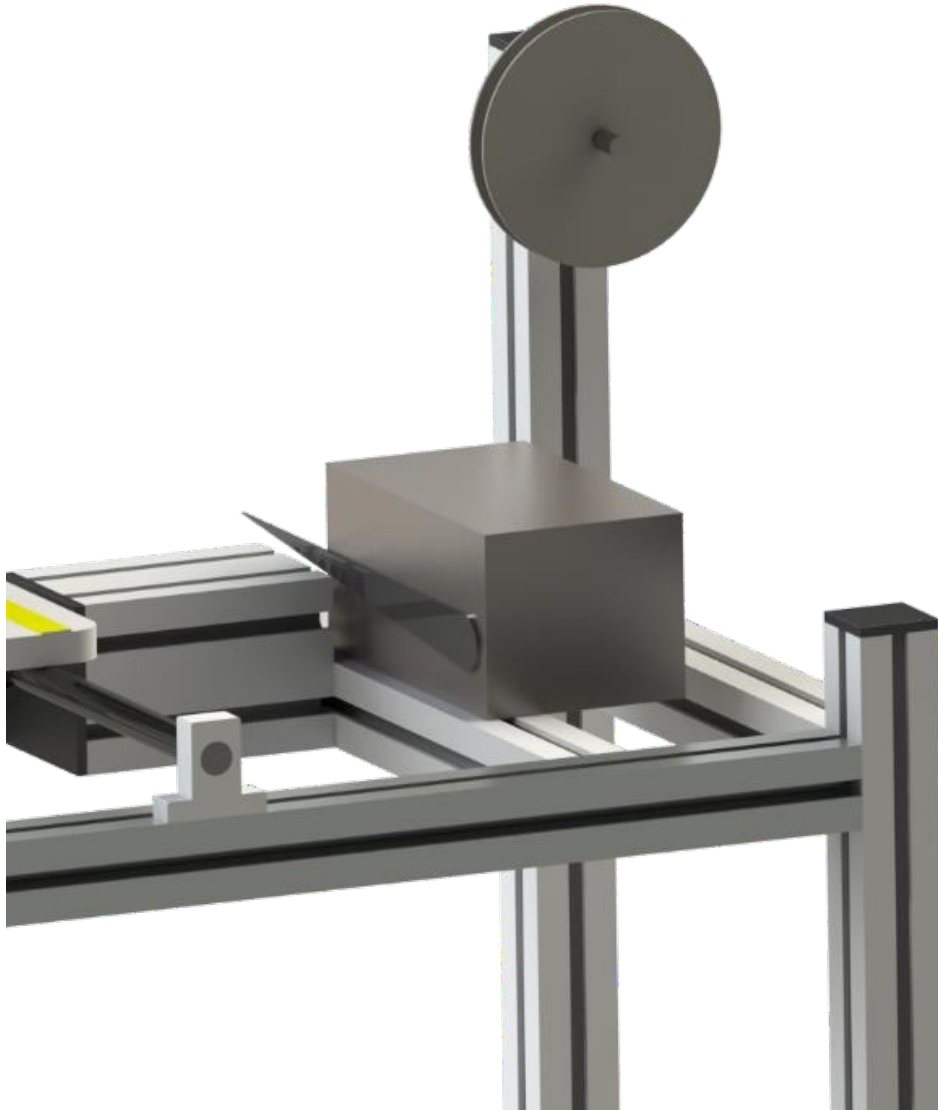
Bonding tape — two-layer structure (transparent + adhesive)



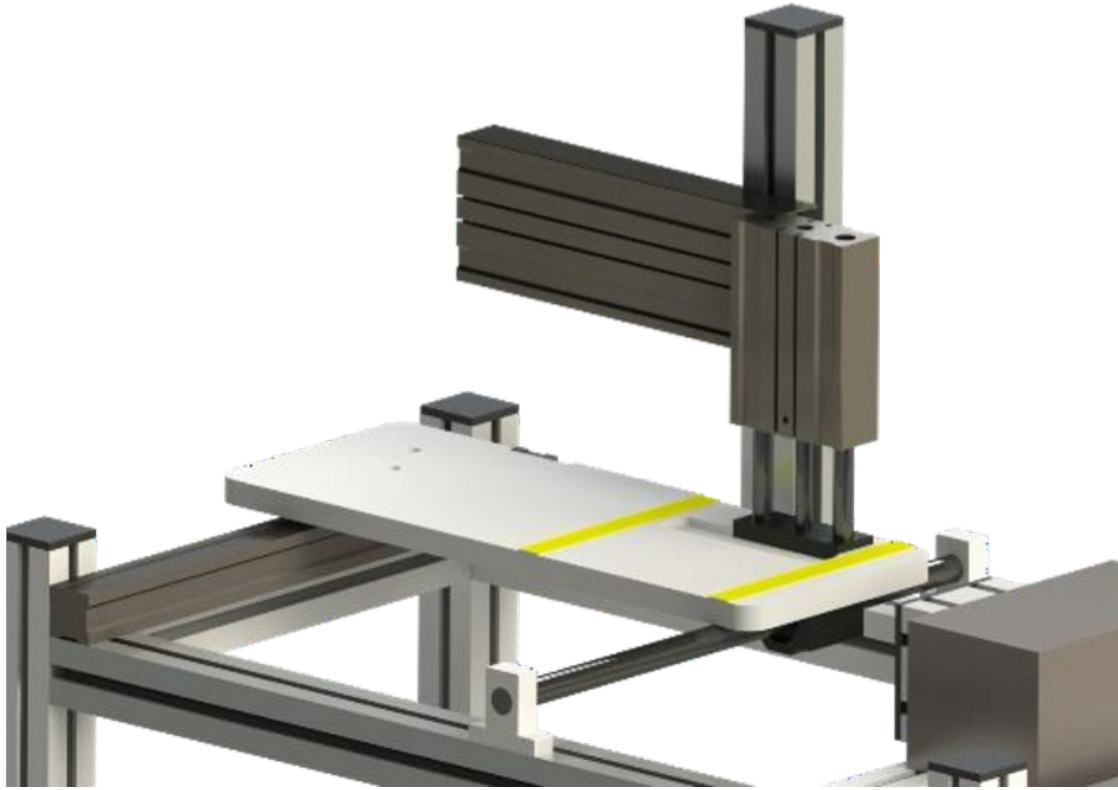
OTTUTHAL machine — front view overview



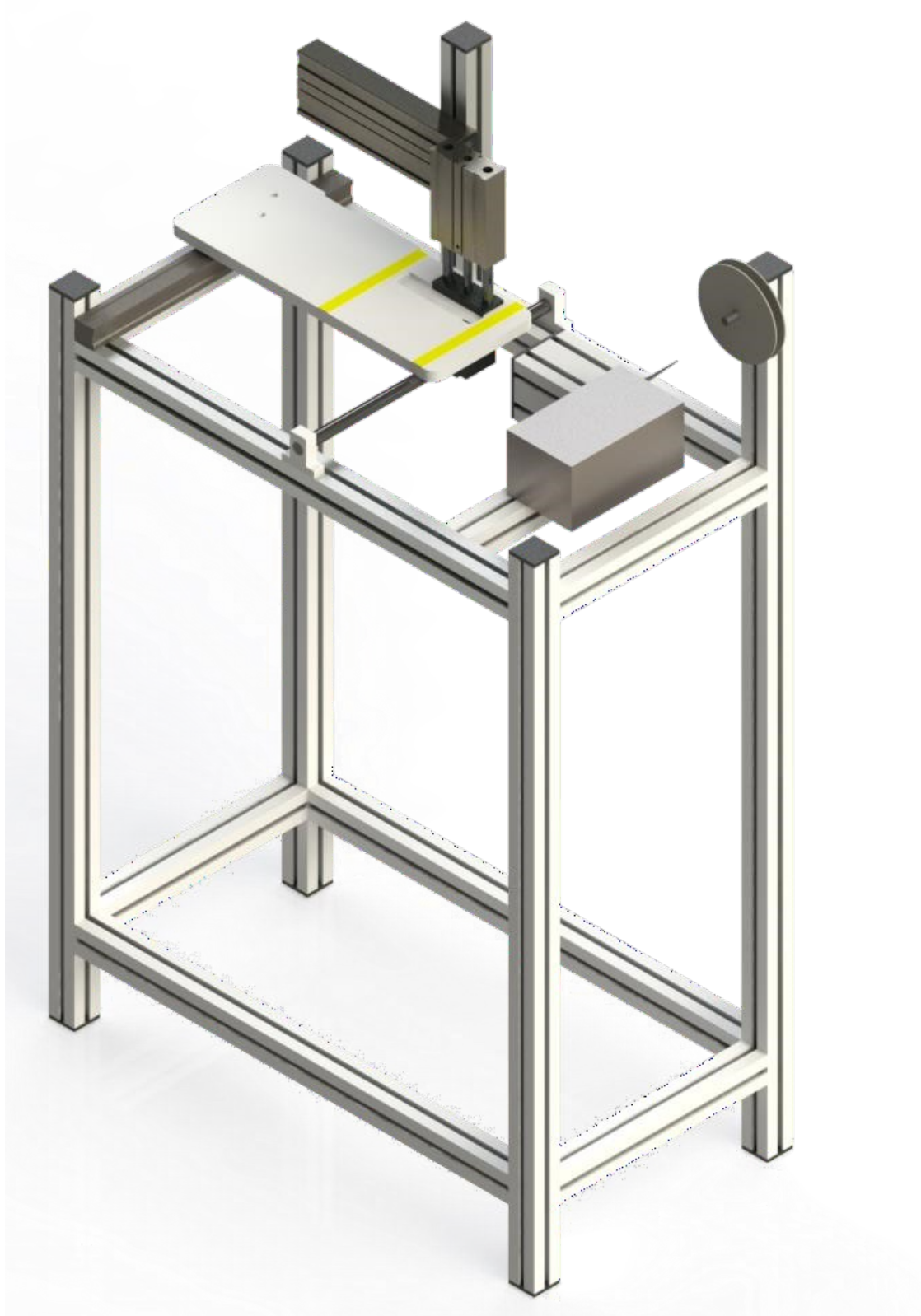
Bonding tape cutting and measuring sub-system



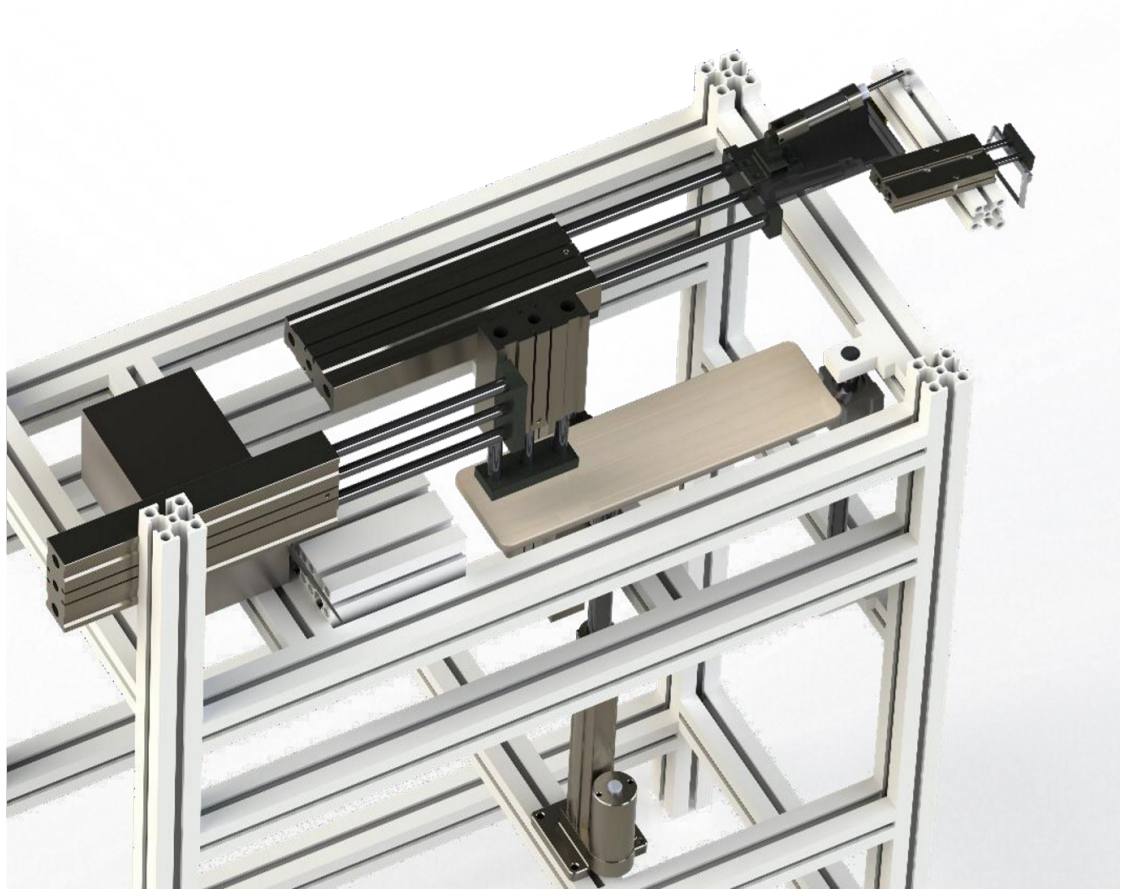
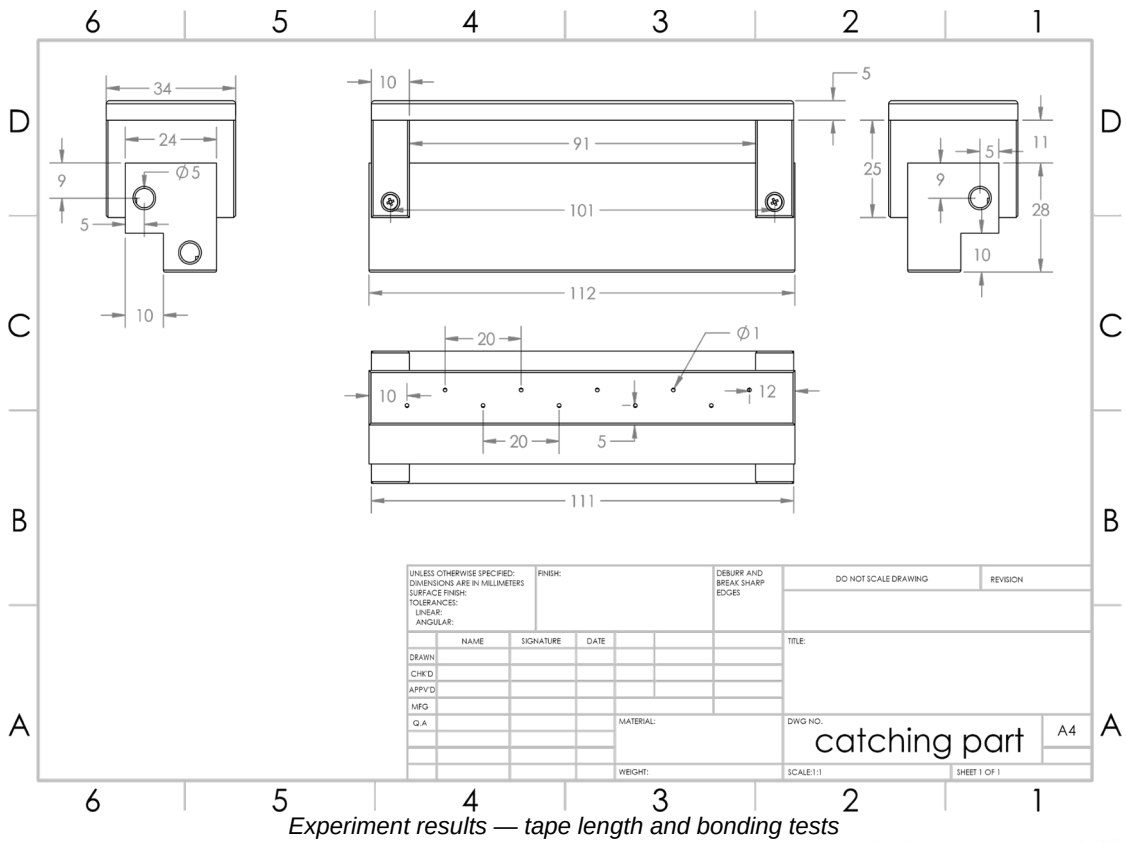
Machine layout — hanging bar, cutter box, and laying stage



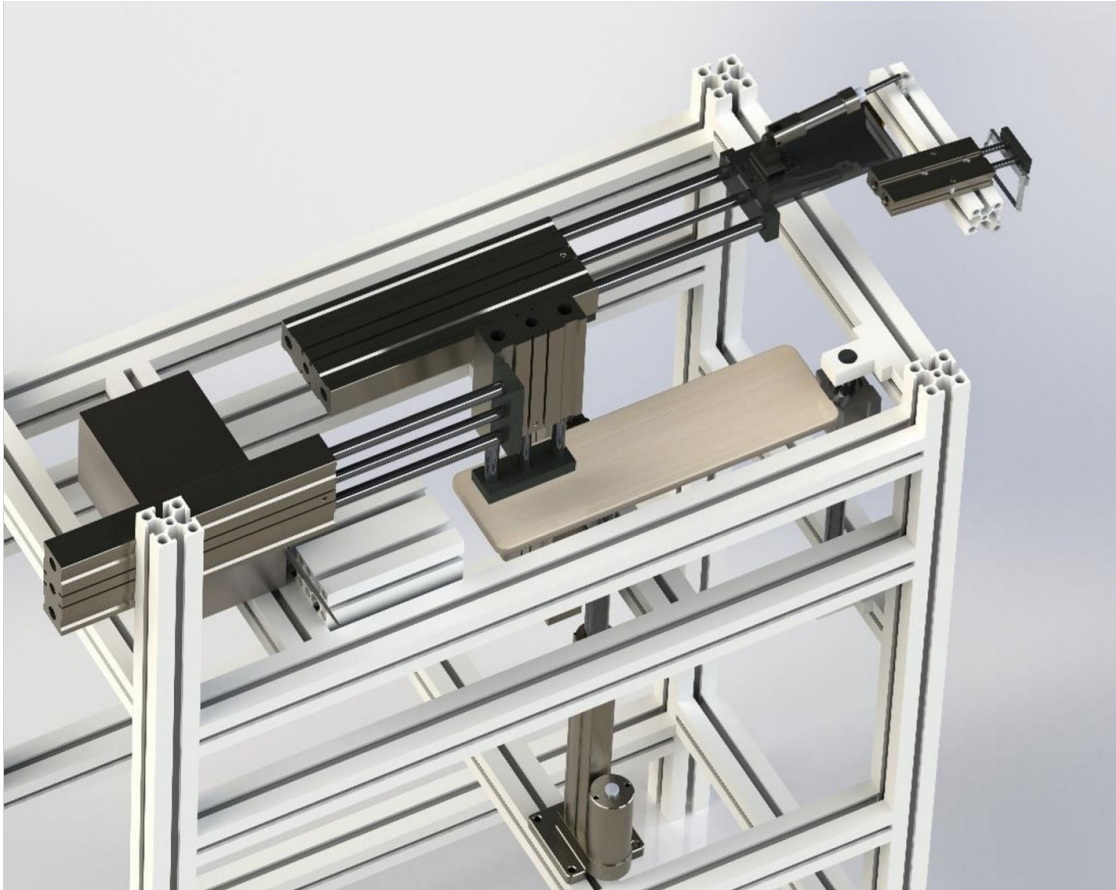
Machine cycle — two-button start and tape positioning sequence



Moving plate returning to initial position after bonding



Recommended fully-automatic machine — CAD drawings



Recommended new machine concept — under development